Supplementary material

Stagnation before exit: performance progression and retention in youth track and field

Anonymised manuscript. Table and section numbers follow the order of first citation in the main text.

Table S1. Sample flow.

Step	n
Athletes born 1998–2006 in the registry extract	32 447
With $\geq$1 approved result at age 13–14	12 938
Also competing at ages 15–16 (landmark condition)	4 910
With known sex	4 909
born 2006 (used only for the activity-at-$\geq$18 outcome)	434
Born 1998–2005	4 475
excluded: no scorable main-group points in both windows	578
excluded: sex $\times$ event-group cells with $<$20 athletes	20
Primary sample (born 1998–2005)	3 877
active seniors	654
with exact birth date (H2 analyses)	3 850
Exit-aligned trajectory subsamples (performance / volume)	1 116 / 1 523
Inclusive best-group exposure sample	3 990
Activity-at-$\geq$18 outcome sample (primary sample plus born 2006)	4 244

The completed STROBE checklist accompanies the submission; the time-stamped analysis plan (committed before estimation) is available with the analysis code. Abbreviation: STROBE, Strengthening the Reporting of Observational Studies in Epidemiology.

Table S2. Robustness of the progression–retention association. OR (95 % CI) per SD of the exposure named (per 100 points for the earlier level given the later level); outcome is senior activity unless stated. Abbreviations: CI, confidence interval; COVID-19, coronavirus disease 2019; GEE, generalised estimating equations; OR, odds ratio; SD, standard deviation; SE, standard error.

Analysis	n	Events	OR (95 % CI)
Primary: Δ mean of 3 best, main event group	3 877	654	2.24 (1.90–2.63)
Exposure: season best (maximum)	3 877	654	2.09 (1.79–2.45)
Exposure: mean of all results	3 877	654	1.68 (1.47–1.93)
Exposure: wind-legal marks only	3 799	652	2.17 (1.85–2.55)
Exposure: equal numbers of results in both windows	3 877	654	1.88 (1.63–2.17)
Exposure: event-equated points	3 877	654	1.94 (1.66–2.26)
Exposure: outdoor marks only	3 403	634	1.98 (1.68–2.33)
Exposure: best event group at 15–16, inclusive sample	3 990	625	2.53 (2.13–3.01)
level and counts from the main group (analysis plan)	3 982	631	1.87 (1.64–2.12)
Splines for result counts and competition days	3 877	654	2.15 (1.80–2.56)
Earlier level given later level, per 100 points	3 877	654	0.75 (0.66–0.85)
Specification check: Δ within ages 13$\rightarrow$14	2 799	565	1.03 (0.91–1.17)
Excluding cohorts born 2004–2005 (COVID window)	2 933	501	2.12 (1.77–2.55)
Progression $\times$ COVID-window interaction	3 877	654	1.30 (0.98–1.74)
$\geq$2 scored results in both windows	3 080	633	2.39 (2.00–2.86)
$\geq$3 scored results in both windows	2 627	609	2.53 (2.08–3.07)
Athletes still competing after age 16	2 001	654	1.83 (1.55–2.17)
Outcome: $\geq$1 result at age $\geq$20	3 877	759	2.00 (1.73–2.31)
Outcome: $\geq$2 results in a year, ages 20–21	3 877	617	2.30 (1.95–2.72)
Outcome: active at $\geq$18 (cohorts incl. 2006)	4 244	1 660	1.55 (1.39–1.72)
Cohort A (born 1998–2001)	1 851	323	2.20 (1.75–2.77)
Cohort B (born 2002–2005)	2 026	331	2.24 (1.77–2.82)
Club-clustered robust SE	3 877	654	2.24 (1.89–2.65)
GEE, exchangeable club correlation	3 877	654	2.22 (1.88–2.63)
E-value for the primary OR (CI bound)			2.36 (2.10)

S3. Deviations from the analysis plan

the H3 interval is a DeLong interval rather than a paired bootstrap; club-grouped cross-validation folds, the exposure defined as residuals against panel growth curves, and a random club intercept were not run (club-clustered standard errors and GEE take the place of the last); the Cox model omits cohort and club size; the cohort replication was run for the H1 odds ratio only, not for H2, H3 or the AUC gain; region, planned as a covariate, is not recorded (see Covariates); and the switcher analysis takes baseline level and result counts from the best event group, where the plan took them from the main group (the planned version is shown alongside in Table S2).